

Minimal Confusable Multisets: A Polynomial-Optimisation Perspective

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June 2026

Abstract

This document takes a step back from the even-odd hypercube construction used in `MinimalConfusableSets` and asks: how might one find the *globally* minimal confusable multiset pair for a given dotting encoder, without any structural assumptions? We reformulate the problem as a polynomial feasibility question, derive an exact algebraic criterion for the smallest non-trivial case ($s = 2$), introduce the sum-of-squares (SOS) / semidefinite programming (SDP) framework for certifying the absence of small confusable pairs, and exhibit an explicit family of size-3 confusable pairs that arises from a cyclic algebraic construction entirely unrelated to the hypercube. Working code is provided in `confusable_search.py` and `confusable_sos.py`.

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1 Problem statement

Fix m encoding directions $\vec{d}_1, \dots, \vec{d}_m \in \mathbb{R}^k$ (the *good bats*). A dotting encoder maps an s -element multiset $X = \{\vec{v}_1, \dots, \vec{v}_s\} \subset \mathbb{R}^k$ to the ms -tuple consisting, for each $\vec{d}_j$, of the n dot products $\vec{v}_i \cdot \vec{d}_j$ sorted in non-decreasing order.

Definition 1. *Two distinct multisets $X, Y \subset \mathbb{R}^k$ of size s are confusable if, for every encoding direction $\vec{d}_j$, the sorted tuples of dot products with $\vec{d}_j$ are identical:*

$$\text{sort}\left(\{\vec{v} \cdot \vec{d}_j : \vec{v} \in X\}\right) = \text{sort}\left(\{\vec{v} \cdot \vec{d}_j : \vec{v} \in Y\}\right).$$

The central question of this document is: for given m, k , and a specific choice of encoding directions, what is the *smallest* s for which a confusable pair of size s exists? We make no assumption about the form that confusable pairs take—in particular, we do not assume they arise from the even-odd hypercube construction.

Power-sum reformulation

Two multisets of real numbers are equal (as multisets) if and only if all their power sums agree. Applied to the scalar projections $\{\vec{v} \cdot \vec{d}_j : \vec{v} \in X\}$, two size- s multisets are confusable if and only if

$$\sum_{\vec{v} \in X} (\vec{v} \cdot \vec{d}_j)^r = \sum_{\vec{v} \in Y} (\vec{v} \cdot \vec{d}_j)^r \quad \text{for all } j = 1, \dots, m \text{ and } r = 1, \dots, s. \quad (1)$$

This is a system of ms polynomial equations in the $2sk$ coordinates of the vectors in X and Y . A confusable pair of size s exists if and only if this system has a solution with $X \neq Y$ as multisets.

Remark 1. *The minimum confusable-pair size is the smallest s for which this polynomial system has a non-trivial solution. It is a property of the specific set of directions $\{\vec{d}_j\}$, not of any particular construction method.*

2 The case $s = 2$: exact characterisation

For $s = 2$ the polynomial system (1) simplifies substantially. Write $X = \{\vec{x}_1, \vec{x}_2\}$ and $Y = \{\vec{y}_1, \vec{y}_2\}$. The $r = 1$ conditions force the centroids to agree: $\vec{x}_1 + \vec{x}_2 = \vec{y}_1 + \vec{y}_2$ (when the $\vec{d}_j$ span $\mathbb{R}^k$). Setting the common centroid to zero (WLOG), write $\vec{x}_i = \pm \vec{p}/2$ and $\vec{y}_i = \pm \vec{q}/2$. The $r = 2$ conditions then reduce to

$$|\vec{p} \cdot \vec{d}_j| = |\vec{q} \cdot \vec{d}_j| \quad \forall j,$$

i.e., for each direction j : either $(\vec{p} + \vec{q}) \cdot \vec{d}_j = 0$ or $(\vec{p} - \vec{q}) \cdot \vec{d}_j = 0$ (or both). Setting $\vec{A} = \vec{p} + \vec{q}$ and $\vec{B} = \vec{p} - \vec{q}$:

Proposition 1. *A confusable pair of size 2 exists if and only if there exist non-zero $\vec{A}, \vec{B} \in \mathbb{R}^k$ with $\vec{A} \neq \pm \vec{B}$ such that every encoding direction satisfies $\vec{d}_j \in \vec{A}^\perp \cup \vec{B}^\perp$.*

In $\mathbb{R}^2$ this says: the m encoding directions can be partitioned into two groups, one whose members are all parallel to a fixed direction $\vec{A}^\perp$, and one whose members are all parallel to $\vec{B}^\perp$. For generic directions (no two parallel), this requires $m \leq 2$.

Worked example

Consider $k = 2$ with two directions $\vec{d}_1 = (1, 0)$ (at 0) and $\vec{d}_2 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ (at 45). Take $\vec{A} \propto (0, 1)$ (so $\vec{A}^\perp$ is the horizontal axis, containing $\vec{d}_1$) and $\vec{B} \propto (1, -1)$ (so $\vec{B}^\perp$ is the 45 axis, containing $\vec{d}_2$). These are non-zero and non-parallel. Then

$$X = \left\{ \frac{1}{4}(1, 1), -\frac{1}{4}(1, 1) \right\}, \quad Y = \left\{ \frac{1}{4}(1, -1), -\frac{1}{4}(1, -1) \right\}$$

is a confusable pair of size 2. One may verify directly: both X and Y project to the sorted tuple $(-\frac{1}{4}, \frac{1}{4})$ onto $\vec{d}_1$, and to $(-\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}})$ onto $\vec{d}_2$.

Adding any third generic direction immediately destroys this: a direction at $\theta \neq 0, 45$ does not lie in $\vec{A}^\perp \cup \vec{B}^\perp$, so Proposition 1 fails, and the SOS certificate in Section 4 confirms injectivity.

3 An explicit $s = 3$ confusable family

A confusable pair of size 3 for the three directions $\vec{d}_1 = (1, 0)$, $\vec{d}_2 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, $\vec{d}_3 = (0, 1)$ can be constructed *without* any reference to the even-odd hypercube. For any triple of distinct real numbers (a, b, c) , define

$$X = \{(a, b), (c, a), (b, c)\}, \quad Y = \{(a, c), (b, a), (c, b)\}.$$

Proposition 2. *For any (a, b, c) with $\{a, b, c\}$ not all equal, the pair (X, Y) defined above is confusable with respect to all three directions $\vec{d}_1, \vec{d}_2, \vec{d}_3$.*

Proof. For $\vec{d}_1 = (1, 0)$: both X and Y have first-coordinates $\{a, c, b\}$ as a multiset.

For $\vec{d}_3 = (0, 1)$: both have second-coordinates $\{b, a, c\}$.

For $\vec{d}_2 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$: the diagonal projections of X are proportional to $\{a+b, c+a, b+c\}$ and those of Y to $\{a+c, b+a, c+b\}$. Since addition is commutative, both sets are $\{a+b, a+c, b+c\}$.

In each case the sorted projections are identical. Since X and Y differ as multisets when a, b, c are not all equal (the first coordinates are the same set but paired with different second coordinates), (X, Y) is a genuine confusable pair. $\square$

Relationship to the even-odd hypercube

The cyclic family has a non-trivial intersection with the $M = 3$ single-cancellation hypercube construction. One can show that a single-cancellation hypercube with $M = 3$ bad bats always produces a centred pair satisfying $EE = -OO$ (the pair is centrosymmetric about its common centroid). The cyclic pair (X, Y) is centrosymmetric if and only if one of a, b, c equals their common mean $\mu = (a + b + c)/3$, i.e. one of the deviations $(a - \mu, b - \mu, c - \mu)$ is zero.

Concretely, the cyclic subfamily with $\{a - \mu, b - \mu, c - \mu\} \propto \{0, 1, -1\}$ is exactly what the standard $M = 3$ hypercube construction produces for the bad bats perpendicular to the three encoding directions.

The remaining members of the cyclic family — those where all three deviations from the mean are non-zero — are genuinely outside the $M = 3$ single-cancellation hypercube. Crucially, this distinction is *translation-invariant*: translating both multisets by the same

vector preserves confusability but leaves the deviations from the mean unchanged, so no translation can convert a non-hypercube instance into a hypercube one.

Remark 2. *Whether the non-hypercube instances of the cyclic family can be produced by a hypercube construction with $M > 3$ or with multiple cancellations is a separate question not pursued here. What is clear is that the full 2-parameter cyclic family is strictly larger than the 1-parameter subfamily that the $M = 3$ single-cancellation hypercube produces, and the two cannot be brought into agreement by any translation.*

For a concrete numerical instance lying outside the hypercube subfamily: $(a, b, c) = (5.5, -5.7, 0.2)$ has mean $\mu = 0$ and deviations $(5.5, -5.7, 0.2)$, none of which is zero. This gives

$$X = \{(5.5, -5.7), (0.2, 5.5), (-5.7, 0.2)\}, \quad Y = \{(5.5, 0.2), (-5.7, 5.5), (0.2, -5.7)\},$$

a confusable pair of size 3 that is confirmed by the numerical code and is not reachable by any translation of any $M = 3$ single-cancellation hypercube pair.

When does a fourth direction destroy confusability?

A direction $\vec{d} = (\cos \theta, \sin \theta)$ with $\theta \neq 0, 45, 90$ will generically break the size-3 confusable pair: the cubic power sum (the $r = 3$ condition) for direction $\vec{d}$ is

$$p_3^X(\theta) - p_3^Y(\theta) = 3 \cos^2 \theta \sin \theta (a^2 b + c^2 a + b^2 c - a^2 c - b^2 a - c^2 b) + 3 \cos \theta \sin^2 \theta (ab^2 + ca^2 + bc^2 - ac^2 - ba^2 - cb^2).$$

This vanishes for all (a, b, c) only when $\cos^2 \theta \sin \theta = \cos \theta \sin^2 \theta$, i.e. $\theta \in \{0, 45, 90\}$. Adding any other direction $\vec{d}_4$ makes the $r = 3$ condition non-trivially constraining, and numerical search (Section 5) finds no confusable pairs of size up to 5 for four equally-spaced directions.

4 Sum-of-squares certificates for size-2 injectivity

The certificate polynomial

For the $s = 2$ question, define

$$f(\vec{A}, \vec{B}) = \sum_{j=1}^m (\vec{A} \cdot \vec{d}_j)^2 (\vec{B} \cdot \vec{d}_j)^2, \quad g(\vec{A}, \vec{B}) = \|\vec{A}\|^2 \|\vec{B}\|^2.$$

From Proposition 1, a confusable pair of size 2 exists if and only if f has a zero for some non-zero $(\vec{A}, \vec{B})$. This is equivalent to f failing to satisfy $f \geq \lambda g$ for any $\lambda > 0$. Conversely:

Observation 1. *If there exists $\lambda > 0$ such that $f - \lambda g$ is a sum-of-squares (SOS) polynomial, then $f(\vec{A}, \vec{B}) \geq \lambda \|\vec{A}\|^2 \|\vec{B}\|^2 > 0$ for all $(\vec{A}, \vec{B}) \neq \vec{0}$, and hence no confusable pair of size 2 exists.*

Note that f itself is already SOS ($f = \sum_j [(\vec{A} \cdot \vec{d}_j)(\vec{B} \cdot \vec{d}_j)]^2$), so $\lambda = 0$ is always feasible. The question is whether the optimal $\lambda^* > 0$.

SDP formulation

An SOS polynomial of degree 4 in $2k$ variables can be written as $v^T Q v$ where v is the vector of all monomials of degree ≤ 2 in the $2k$ variables $(\vec{A}, \vec{B})$ and Q is a PSD (positive semidefinite) matrix. The number of such monomials is $\binom{2k+2}{2}$, giving a $\binom{2k+2}{2} \times \binom{2k+2}{2}$ matrix. For $k = 2$: a 15×15 matrix.

The SDP is:

$$\begin{aligned} & \text{maximise} \quad \lambda \\ & \text{subject to} \quad Q \succeq 0, \\ & \quad \quad v^T Q v = f - \lambda g \quad (\text{as polynomial identity}). \end{aligned}$$

The polynomial identity is enforced by matching the coefficient of each degree-4 monomial z^γ on both sides:

$$\sum_{\substack{\alpha+\beta=\gamma \\ |\alpha|, |\beta| \leq 2}} Q_{\alpha\beta} = f_\gamma - \lambda g_\gamma,$$

where f_γ and g_γ are the known coefficients of f and g at monomial γ . These are linear equalities in the decision variables (Q, λ) , so the overall problem is a standard SDP.

Results for $k = 2$

Running the SDP (implemented in `confusable_sos.py`) for three choices of encoding directions in $\mathbb{R}^2$:

Directions	λ^*	Conclusion
$\vec{d}_1 = (1, 0)$, $\vec{d}_2 = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ (2 dirs)	0.000	Confusable size-2 pair may exist
Add $\vec{d}_3 = (0, 1)$ (3 dirs)	0.167	Certified: no confusable size-2 pair
Add $\vec{d}_4 = (-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ (4 dirs)	0.500	Certified: no confusable size-2 pair

For 2 directions, $\lambda^* = 0$ confirms that a confusable pair exists (as constructed in Section 2). For 3 and 4 directions, $\lambda^* > 0$ provides a rigorous certificate of injectivity for size-2 multisets.

5 Numerical search for larger s

For $s \geq 3$, the polynomial system (1) is harder to analyse algebraically, and the SDP certificate needs to be extended to higher-dimensional moment matrices (see Section 6). A practical complement is a gradient-based numerical search.

Search strategy

For fixed s, k , and directions D , the code in `confusable_search.py` proceeds as follows:

1. Sample $X \in \mathbb{R}^{s \times k}$ at random with zero centroid.
2. Compute the target moments $t = (p_{r,j}(X))_{r,j}$ via equation (1).

3. From a *random* starting point Y_0 (not near X), minimise the confusability loss $\|p(Y) - t\|^2$ over Y using L-BFGS-B.
4. If the minimised loss falls below 10^{-10} and the Hungarian matching distance between X and Y exceeds 10^{-2} , report a confusable pair.

The random initialisation of Y_0 is critical: starting near X causes the optimiser to converge trivially to $Y = X$.

Results for $k = 2$

Running 500 random trials for each (s, m) pair:

s	$m = 2$ dirs	$m = 3$ dirs	$m = 4$ dirs
2	Found	not found	not found
3	Found	Found	not found
4	Found	Found	not found
5	Found	Found	not found

With only 2 directions confusable pairs are found for every $s \geq 2$, as expected (two directions generically leave a 1-dimensional family of confusable pairs at each size). With 3 directions confusable pairs exist for $s \geq 3$ (including the cyclic family of Section 3). With 4 equally-spaced directions, no confusable pair was found for any $s \leq 5$ in 500 trials, which gives numerical evidence in support of the conjecture that 4 directions suffice to inject all multisets of size ≤ 5 of 2-vectors.

Remark 3. *Failure to find a confusable pair by this search is not a proof of injectivity: it may simply mean that confusable pairs exist but are rare or have small basins of attraction. A rigorous lower bound requires the Lasserre hierarchy (Section 6).*

6 The Lasserre hierarchy for general s

Moment relaxation

The existence of a confusable pair of size s is equivalent to the feasibility of the polynomial system (1) with $X \neq Y$. The *Lasserre moment hierarchy* provides a sequence of SDPs, parametrised by a *relaxation level* $d = 1, 2, \dots$, that gives increasingly tight approximations to this feasibility question.

At level d , the decision variables are the *moment variables* y_α for all multi-indices α with $|\alpha| \leq 2d$, where y_α represents the expectation $\mathbb{E}[z^\alpha]$ of monomial z^α under a hypothetical probability measure supported on the feasible set. The SDP constraints are:

- *Moment matrix:* $M_d(y) \succeq 0$ (a $\binom{n+d}{d} \times \binom{n+d}{d}$ PSD matrix, where $n = 2sk$ is the number of scalar decision variables in X and Y).
- *Constraint moments:* $\sum_\alpha c_\alpha y_\alpha = 0$ for each equality constraint in (1) (linear in y).

If this SDP is infeasible at level d , then no confusable pair of size s exists. If it is feasible, the solution can sometimes be *flat-extended* to recover an explicit confusable pair.

Complexity

The moment matrix at level d has $\binom{n+d}{d}$ rows and columns. For $k = 2$, $s = 4$: $n = 16$, and at $d = 2$ the moment matrix is $\binom{18}{2} = 153 \times 153$. This is tractable with a modern SDP solver (MOSEK, SCS, or Clarabel). For $s = 5$: $n = 20$, moment matrix $\binom{22}{2} = 231 \times 231$ at $d = 2$.

A practical implementation in Python would use `cvxpy` to construct the moment matrix and add the confusability constraints as linear equality constraints on the moments. The `confusable_sos.py` file implements the $s = 2$ case directly; extending it to general s follows the same pattern with a larger moment matrix.

7 Code notes

Two companion Python files are provided, both requiring `numpy`, `scipy`, `sympy`, and `cvxpy` (`pip install cvxpy sympy`).

`confusable_search.py`

Implements the gradient-based search described in Section 5. Main entry point:

```
result = search_confusable_pair(s, k, directions,
                                n_restarts=500, res_tol=1e-10)
# Returns (X, Y, residual) if found, None otherwise.
```

The function accepts any s , k , and list of direction vectors. Running the script directly exercises the cases in the table above.

`confusable_sos.py`

Implements:

- `sos_lower_bound(directions)`: the SDP from Section 4, maximising λ s.t. $f - \lambda g$ is SOS. Returns λ^* . If $\lambda^* > 0$, no confusable size-2 pair exists.
- `exact_check_s2(directions)`: an exhaustive $O(2^m)$ combinatorial check of Proposition 1 for small m . Returns `(True, A, B, X, Y)` if a confusable pair is found.

```
lam = sos_lower_bound(directions) # SDP certificate
found, A, B, X, Y = exact_check_s2(directions) # Combinatorial
check
```